

Calculus III — Quiz 3, everything

LESSONS 5–7 · §14.1 VECTOR FUNCTIONS · §14.2 DERIVATIVES & INTEGRALS · §14.3 MOTION

Read §1 and §2. Do the problems. Everything here is verified; answers were recomputed, not asserted.

1. The whole quiz on one page

If you read nothing else, read this. Every formula below is derived in §2 — so if you forget one, you can rebuild it rather than guess.

The single idea. A vector function is three ordinary functions wearing one coat. *Everything* — limits, derivatives, integrals, continuity — is done **slot by slot** and then reassembled. There is no new calculus in this quiz.

Position, velocity, acceleration

$$\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$$

$$\vec{v} = \vec{r}' = \langle x', y', z' \rangle \quad \vec{a} = \vec{r}''$$

Speed is a number, velocity is a vector

$$\text{speed} = |\vec{v}(t)| = \sqrt{x'^2 + y'^2 + z'^2}$$

Unit tangent — direction only, length 1

$$\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$$

Arc length — rate $\times$ time, summed

$$L = \int_a^b |\vec{r}'(t)| dt$$

Integrals — slot by slot, vector constant

$$\int \vec{r} dt = \left\langle \int x, \int y, \int z \right\rangle + \vec{C}$$

$\vec{C}$ has **three** components; $\vec{r}(0)$ finds it.

Limits and continuity — slot by slot

$$\lim_{t \rightarrow a} \vec{r}(t) = \langle \lim x, \lim y, \lim z \rangle$$

Domain = where *every* slot is defined at once.

Projectile — $\vec{a} = \langle 0, 0, -9.8 \rangle$

$$\vec{r}(t) = \langle 0, 0, -4.9t^2 \rangle + t \vec{v}(0) + \vec{r}(0)$$

Product rules — order matters for $\times$

$$(\vec{u} \cdot \vec{v})' = \vec{u}' \cdot \vec{v} + \vec{u} \cdot \vec{v}'$$

$$(\vec{u} \times \vec{v})' = \vec{u}' \times \vec{v} + \vec{u} \times \vec{v}'$$

THE MARK YOU WILL LOSE “Find the **speed**” means take the square root at the end. If your answer to a speed question still has $\langle \rangle$ around it, you stopped one step early.

Domains, in one line each. $\sqrt{\quad}$ needs ≥ 0 · $\ln$ needs > 0 (strict) · denominators need $\neq 0$ · e^t , $\sin$, $\cos$ are fine everywhere.
Then **intersect** the conditions — every slot must work at the same time.

2. Build it up — so you can rebuild it

Why everything goes slot by slot

Between time t and a moment h later the bug moves from $\vec{r}(t)$ to $\vec{r}(t+h)$. The change is the subtraction, and subtracting triples means subtracting each slot:

$$\vec{r}(t+h) - \vec{r}(t) = \langle x(t+h) - x(t), y(t+h) - y(t), z(t+h) - z(t) \rangle$$

Dividing by h divides each slot, so each becomes its own rise-over-run. Letting $h \rightarrow 0$, a point is close to another exactly when it is close in *all three* coordinates — so the limit of a triple is the triple of the limits.

$$\vec{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$$

Slot by slot is forced, not a convention. The slots never had an opportunity to interact. The same argument covers limits, integrals and continuity, because all of them are built from subtracting, dividing and taking limits.

Why $\vec{r}'$ points along the curve

$\vec{r}(t+h) - \vec{r}(t)$ is the arrow from where you are to where you will be — a chord cutting across the curve. Dividing by a positive h stretches it but **cannot rotate it**, so the direction survives. As h shrinks the far end slides back and the

chord pivots until it grazes the curve. Hence $\vec{r}'$ points exactly the way you are heading, and its length says how fast.

Why speed is $|\vec{r}'|$, and arc length for free

$|\vec{r}(t+h) - \vec{r}(t)|$ is the distance travelled in h seconds; divide by h and you have distance over time. In the limit that is the speed. And if you move at speed $|\vec{r}'|$ for a tiny time dt you cover $|\vec{r}'| dt$ of ground — add up every piece and you have arc length. **It is distance = rate $\times$ time, summed.** Rebuild it from that sentence if you blank.

Why acceleration leans into the bend

Suppose speed never changes. Then $|\vec{v}|^2 = \vec{v} \cdot \vec{v}$ is constant, so differentiating gives

$$0 = \frac{d}{dt}(\vec{v} \cdot \vec{v}) = \vec{v}' \cdot \vec{v} + \vec{v} \cdot \vec{v}' = 2\vec{a} \cdot \vec{v} \implies \vec{a} \perp \vec{v}.$$

Physically: acceleration along your direction of travel would speed you up or slow you down. If speed is constant, all of it is spent **turning**. For $\vec{r} = \langle a \cos t, a \sin t \rangle$ you get $\vec{a} = -\vec{r}$ — pointing straight back at the centre.

Projectiles are Newton plus those integrals

Gravity is a constant acceleration and nothing pushes sideways, so $\vec{a} = \langle 0, 0, -9.8 \rangle$. Integrate once for $\vec{v}$, again for $\vec{r}$, using $\vec{v}(0)$ and $\vec{r}(0)$ to fix each vector constant. Then **every question is the z slot**, because only z has gravity in it.

3. The algorithms — eight recipes

1. Domain of a vector function

- Write the condition each slot needs: $\sqrt{\quad} \geq 0$, $\ln > 0$, denominator $\neq 0$.
- Solve each one separately as an interval.
- Intersect** them. All slots must work at once.
- Watch bracket types: $\sqrt{\quad}$ allows equality, $\ln$ never does.

2. Limit of a vector function

- Do each slot as a separate one-variable limit.
- Recognise the standard ones: $\frac{\sin kt}{t} \rightarrow k$, $\frac{e^t - 1}{t} \rightarrow 1$, $\frac{1 - \cos t}{t^2} \rightarrow \frac{1}{2}$.
- Bounded $\times$ something going to zero $\rightarrow$ zero (squeeze).
- Reassemble into $\langle \quad, \quad, \quad \rangle$.

3. Velocity, speed, acceleration

- Differentiate each slot $\rightarrow \vec{v}$.
- Differentiate again $\rightarrow \vec{a}$.
- Substitute the time *after* differentiating, never before.
- If it asked for **speed**, take $|\vec{v}| = \sqrt{\text{sum of squares}}$.

4. Unit tangent $\vec{T}$

- $\vec{r}'(t)$, then substitute the time.
- Compute $|\vec{r}'|$ at that time.
- Divide the vector by that number.
- Sanity check: the components squared must sum to 1.

5. Position from velocity or acceleration

- Integrate slot by slot, adding $\vec{C} = \langle C_1, C_2, C_3 \rangle$.
- Substitute $t = 0$ — every t term dies, so what is left *is* $\vec{C}$.
- Match against the given initial position/velocity to read off all three.
- Integrate again if you started from acceleration.

6. Tangent line to a curve at t_0

- Point: $\vec{r}(t_0)$. Direction: $\vec{r}'(t_0)$.
- Line: $\vec{L}(u) = \vec{r}(t_0) + u\vec{r}'(t_0)$ — **a new parameter u** .
- To hit a plane, set that coordinate to 0 and solve for u .
- Substitute u back to get the point.

7. Projectile

- $\vec{a} = \langle 0, 0, -9.8 \rangle$. Integrate $\rightarrow \vec{v}$, add $\vec{v}(0)$.
- Integrate $\rightarrow \vec{r}$, add $\vec{r}(0)$.
- Max height**: solve $z' = 0$, put that t into z .
- Time of flight**: solve $z = 0$. **Range**: that t into x, y .

8. What curve is this?

- Look for $\cos^2 + \sin^2$: that is a circle of radius a in some plane.
- A slot that is constant $\Rightarrow$ the curve is flat in that direction.
- A slot linear in t while the others go round $\Rightarrow$ helix.
- Eliminate t between two slots to get the surface it lies on.

4. Every official problem, worked

These are the department's own Exam 1 study-guide problems for Lessons 5–7 — the closest thing to the real quiz that exists. Cover the solution, try it, then check.

Vector-Valued Functions §14.1

1. Find the domain on which $\vec{r}(t) = \langle \sqrt{t-1}, \ln(4-t), e^t \rangle$ is continuous.

Answer: $[1, 4)$

1. A vector function is continuous exactly where **every** component is.
2. $\sqrt{t-1}$ needs $t \geq 1$.
3. $\ln(4-t)$ needs $4-t > 0$, i.e. $t < 4$.
4. e^t is continuous everywhere. Intersect: $[1, 4)$.

TRAP The bracket types. $\sqrt{}$ allows equality (closed at 1); $\ln$ never does (open at 4).

2. Evaluate $\lim_{t \rightarrow 0} \left\langle \frac{\sin(3t)}{t}, e^{2t}, \frac{t^2+1}{t+1} \right\rangle$.

Answer: $\langle 3, 1, 1 \rangle$

1. Componentwise.
2. $\lim_{t \rightarrow 0} \frac{\sin 3t}{t} = 3$ (the $\frac{\sin u}{u}$ limit with $u = 3t$).
3. $\lim_{t \rightarrow 0} e^{2t} = 1$.
4. $\lim_{t \rightarrow 0} \frac{t^2+1}{t+1} = 1$. So the limit is $\langle 3, 1, 1 \rangle$.

TRAP Writing 1 for the first component out of reflex. $\frac{\sin 3t}{t} = 3 \cdot \frac{\sin 3t}{3t} \rightarrow 3$.

3. Find the domain on which $\vec{r}(t) = \left\langle \frac{1}{t^2-4}, \sqrt{t}, \cos(t) \right\rangle$ is continuous.

Answer: $[0, 2) \cup (2, \infty)$

1. $\frac{1}{t^2-4}$ needs $t \neq \pm 2$.
2. $\sqrt{t}$ needs $t \geq 0$ — which already kills $t = -2$.
3. $\cos t$ is everywhere continuous.
4. $[0, \infty)$ with $t = 2$ removed: $[0, 2) \cup (2, \infty)$.

TRAP Reporting both ± 2 as holes. $t = -2$ is not a hole, it is outside the domain the square root already imposed.

4. Describe the curve traced by $\vec{r}(t) = \langle 2 \cos t, 2 \sin t, 0 \rangle$ for $0 \leq t \leq 2\pi$.

Answer: Circle of radius 2 in the xy -plane centered at the origin

1. $x = 2 \cos t, y = 2 \sin t$ gives $x^2 + y^2 = 4$, and $z = 0$ throughout.
2. As t runs $0 \rightarrow 2\pi$ the point goes once around, counterclockwise from $(2, 0)$.
3. **A circle of radius 2 in the xy -plane, centred at the origin.**

TRAP Calling it a helix. A helix needs z to move; here $z \equiv 0$.

5. Evaluate $\lim_{t \rightarrow 0} \left\langle \frac{e^t - 1}{t}, \frac{1 - \cos t}{t^2}, t \sin\left(\frac{1}{t}\right) \right\rangle$.

Answer: $\left\langle 1, \frac{1}{2}, 0 \right\rangle$

1. $\lim_{t \rightarrow 0} \frac{e^t - 1}{t} = 1$ — this is $\frac{d}{dt} e^t$ at 0.
2. $\lim_{t \rightarrow 0} \frac{1 - \cos t}{t^2} = \frac{1}{2}$ (two applications of l'Hôpital, or the series $1 - \cos t \approx \frac{t^2}{2}$).
3. $|t \sin \frac{1}{t}| \leq |t| \rightarrow 0$, so the third component is 0 by the squeeze.
4. $\langle 1, \frac{1}{2}, 0 \rangle$.

TRAP Declaring the third component undefined because $\sin \frac{1}{t}$ oscillates. Bounded times something going to zero still goes to zero.

Calculus of Vector-Valued Functions & Motion in Space §14.2–14.3

6. A particle has velocity $\vec{v}(t) = \langle 2t, 3, -1 \rangle$ and at $t = 0$ passes through $(1, -2, 5)$. Find $\vec{r}(t)$.

Answer: $\vec{r}(t) = \langle t^2 + 1, 3t - 2, -t + 5 \rangle$

1. Position is the antiderivative of velocity: $\vec{r}(t) = \int \vec{v}(t) dt + \vec{C}$.
2. $\int \langle 2t, 3, -1 \rangle dt = \langle t^2, 3t, -t \rangle + \vec{C}$.
3. $\vec{r}(0) = \vec{C} = \langle 1, -2, 5 \rangle$.
4. $\vec{r}(t) = \langle t^2 + 1, 3t - 2, -t + 5 \rangle$.

TRAP Forgetting that $\vec{C}$ is a **vector** of three separate constants, not one scalar.

7. The tangent line to $\vec{r}(t) = \langle t^2, 2t + 1, t^3 \rangle$ at $t = 1$ intersects the xz -plane. Find the point of intersection.

Answer: $\left(-2, 0, -\frac{7}{2}\right)$

1. $\vec{r}(1) = \langle 1, 3, 1 \rangle$ and $\vec{r}'(t) = \langle 2t, 2, 3t^2 \rangle$, so $\vec{r}'(1) = \langle 2, 2, 3 \rangle$.
2. Tangent line: $\langle 1 + 2u, 3 + 2u, 1 + 3u \rangle$.
3. The xz -plane is $y = 0$: $3 + 2u = 0 \Rightarrow u = -\frac{3}{2}$.
4. $x = 1 + 2(-\frac{3}{2}) = -2$, $z = 1 + 3(-\frac{3}{2}) = -\frac{7}{2}$. Point: $(-2, 0, -\frac{7}{2})$.

TRAP Using t again for the tangent line. The line has its own parameter; reusing t makes you solve on the curve instead of on its tangent.

8. Find the unit tangent vector $\vec{T}(0)$ for $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$.

Answer: $\vec{T}(0) = \left\langle 0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$

1. $\vec{r}'(t) = \langle -\sin t, \cos t, 1 \rangle$, so $\vec{r}'(0) = \langle 0, 1, 1 \rangle$.
2. $|\vec{r}'(0)| = \sqrt{0 + 1 + 1} = \sqrt{2}$.
3. $\vec{T}(0) = \left\langle 0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$.

TRAP Skipping the normalisation. $\vec{T}$ is a **unit** vector by definition; $\vec{r}'$ alone is not one here.

9. Find the speed of a particle with position $\vec{r}(t) = \langle e^t, e^{-t}, \sqrt{2}t \rangle$ at $t = 0$.

Answer: Speed = 2

1. $\vec{r}'(t) = \langle e^t, -e^{-t}, \sqrt{2} \rangle$; at $t = 0$ this is $\langle 1, -1, \sqrt{2} \rangle$.
2. Speed = $\sqrt{1 + 1 + 2} = \sqrt{4} = 2$.

TRAP Differentiating e^{-t} to e^{-t} . The chain rule puts a minus in front, though here it squares away — the sign matters for direction, not for speed.

10. If $\vec{r}(t) = \langle t, t^2, t^3 \rangle$, compute $\vec{r}'(t) \times \vec{r}''(t)$.

Answer: $\langle 6t^2, -6t, 2 \rangle$

1. $\vec{r}' = \langle 1, 2t, 3t^2 \rangle$ and $\vec{r}'' = \langle 0, 2, 6t \rangle$.
2. $\vec{i}$: $(2t)(6t) - (3t^2)(2) = 12t^2 - 6t^2 = 6t^2$.
3. $\vec{j}$: $-[(1)(6t) - (3t^2)(0)] = -6t$.
4. $\vec{k}$: $(1)(2) - (2t)(0) = 2$. So $\vec{r}' \times \vec{r}'' = \langle 6t^2, -6t, 2 \rangle$.

TRAP This exact vector is the numerator of the curvature formula $\kappa = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}$ — recognise it and Lesson 8 gets shorter.

11. A particle travels along $\vec{r}(t) = \langle 3 \cos t, 4 \sin t, 0 \rangle$. Find the speed at $t = \frac{\pi}{2}$.

Answer: Speed = 3

1. $\vec{r}'(t) = \langle -3 \sin t, 4 \cos t, 0 \rangle$.
2. At $t = \frac{\pi}{2}$: $\langle -3, 0, 0 \rangle$.
3. Speed = 3.

TRAP Assuming an ellipse is traversed at constant speed. It is not — the speed here is $\sqrt{9 \sin^2 t + 16 \cos^2 t}$, which runs between 3 and 4. Only the circle has constant speed under this parametrisation.

5. More practice — multiple choice

Same sections, different numbers. Answers and full solutions start on the next section, so you can work these honestly.

1. 14.1 · Domain of a vector function

Find the domain of $\mathbf{r}(t) = \left\langle \sqrt{t-1}, \ln(4-t), \frac{1}{t} \right\rangle$.

- (A) $[1, 4)$
- (B) $(1, 4)$
- (C) $[1, 4]$
- (D) $(-\infty, 4)$
- (E) $[1, \infty)$
- (F) $(0, 4)$

2. 14.1 · Circle at a given height

Which function traces the circle of radius 3 centred on the z -axis in the plane $z = 5$, counterclockwise as seen from above?

- (A) $\mathbf{r}(t) = \langle 3 \cos t, 3 \sin t, 5 \rangle$
- (B) $\mathbf{r}(t) = \langle 3 \cos t, 3 \sin t, t \rangle$
- (C) $\mathbf{r}(t) = \langle 5 \cos t, 5 \sin t, 3 \rangle$
- (D) $\mathbf{r}(t) = \langle 3 \cos t, -3 \sin t, 5 \rangle$
- (E) $\mathbf{r}(t) = \langle 3 + \cos t, 3 + \sin t, 5 \rangle$
- (F) $\mathbf{r}(t) = \langle 9 \cos t, 9 \sin t, 5 \rangle$

3. 14.1 · The surface a curve lies on

On which surface does $\mathbf{r}(t) = \langle t \cos t, t \sin t, t \rangle$ lie for $t \geq 0$?

- (A) the cone $x^2 + y^2 = z^2$
- (B) the cylinder $x^2 + y^2 = 1$
- (C) the paraboloid $z = x^2 + y^2$
- (D) the sphere $x^2 + y^2 + z^2 = 1$
- (E) the plane $z = 0$
- (F) the hyperboloid $x^2 + y^2 - z^2 = 1$

4. 14.1 · Intersection with a surface

The helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ meets the sphere $x^2 + y^2 + z^2 = 5$. At which $t \geq 0$?

- (A) $t = 2$
- (B) $t = \sqrt{5}$
- (C) $t = 4$
- (D) $t = 2\pi$
- (E) $t = \sqrt{2}$
- (F) $t = 5$

5. 14.2 · Derivative of a vector function

For $\mathbf{r}(t) = \langle t^2, e^{2t}, \ln t \rangle$, find $\mathbf{r}'(1)$.

- (A) $\langle 2, 2e^2, 1 \rangle$
- (B) $\langle 2, e^2, 1 \rangle$
- (C) $\langle 2, 2e^2, 0 \rangle$
- (D) $\langle 1, 2e^2, 1 \rangle$
- (E) $\langle 2, 2e, 1 \rangle$
- (F) $\langle 2, e^2, 0 \rangle$

6. 14.2 · Tangent line to a curve

Find parametric equations for the tangent line to $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ at the point $(2, 4, 8)$.

- (A) $x = 2 + s, y = 4 + 4s, z = 8 + 12s$
- (B) $x = 1 + 2s, y = 4 + 4s, z = 12 + 8s$
- (C) $x = 2 + s, y = 4 + 2s, z = 8 + 3s$
- (D) $x = 2 + 2s, y = 4 + 4s, z = 8 + 8s$
- (E) $x = s, y = 4s, z = 12s$
- (F) $x = 2 + s, y = 4 + 4s, z = 8 + 6s$

7. 14.2 · Derivative rules

$\mathbf{r}(t)$ is differentiable and $|\mathbf{r}(t)|$ is **constant**. What is $\mathbf{r}(t) \cdot \mathbf{r}'(t)$?

- (A) 0
- (B) $|\mathbf{r}(t)|$
- (C) 1
- (D) $|\mathbf{r}'(t)|^2$

- (E) $2|\mathbf{r}(t)||\mathbf{r}'(t)|$
- (F) it cannot be determined

8. 14.2 · *Definite integral*

Evaluate $\int_0^1 \langle 6t^2, 4t, e^t \rangle dt$.

- (A) $\langle 2, 2, e - 1 \rangle$
- (B) $\langle 2, 2, e \rangle$
- (C) $\langle 6, 4, e - 1 \rangle$
- (D) $\langle 2, 2, 1 - e \rangle$
- (E) $\langle 3, 2, e - 1 \rangle$
- (F) $\langle 2, 4, e - 1 \rangle$

9. 14.3 · *Velocity, speed, acceleration*

A particle has position $\mathbf{r}(t) = \langle 3t, t^2, t^3 \rangle$. Find its **speed** at $t = 1$.

- (A) $\sqrt{22}$
- (B) $\sqrt{14}$
- (C) 22
- (D) $\sqrt{13}$
- (E) $\sqrt{34}$
- (F) 4

10. 14.3 · *Circular motion*

For $\mathbf{r}(t) = \langle 4 \cos 3t, 4 \sin 3t \rangle$, which statement is true?

- (A) the speed is constant at 12, and the acceleration points toward the origin
- (B) the speed is constant at 4, and the acceleration is zero
- (C) the speed is constant at 3, and the acceleration points away from the origin
- (D) the speed increases with t
- (E) the acceleration is parallel to the velocity
- (F) the speed is constant at 12, and the acceleration is zero

11. 14.3 · *Position from acceleration*

A particle has acceleration $\mathbf{a}(t) = \langle 0, 2, 6t \rangle$, with $\mathbf{v}(0) = \langle 1, 0, 0 \rangle$ and $\mathbf{r}(0) = \langle 0, 1, 2 \rangle$. Find $\mathbf{r}(1)$.

- (A) $\langle 1, 2, 3 \rangle$
- (B) $\langle 1, 1, 2 \rangle$
- (C) $\langle 1, 3, 5 \rangle$
- (D) $\langle 0, 2, 3 \rangle$
- (E) $\langle 1, 2, 5 \rangle$
- (F) $\langle 2, 2, 3 \rangle$

12. 14.3 · *Projectile · time of flight*

A ball is launched from a height of 15 m with initial velocity $\langle 20, 0, 10 \rangle$ m/s, under acceleration $\langle 0, 0, -10 \rangle$ m/s². When does it hit the ground?

- (A) $t = 3$ s
- (B) $t = 2$ s
- (C) $t = 1$ s
- (D) $t = 1.5$ s
- (E) $t = 4$ s
- (F) $t = 5$ s

13. 14.3 · *Projectile · maximum height*

Same ball: launched from 15 m with velocity $\langle 20, 0, 10 \rangle$ m/s and acceleration $\langle 0, 0, -10 \rangle$ m/s². What is its maximum height?

- (A) 20 m
- (B) 15 m
- (C) 25 m
- (D) 10 m
- (E) 30 m
- (F) 5 m

6. Answers and solutions

Each one restates the problem, then works straight down. Read it as a worked example of its type — the heading names the type, so the same steps carry to every question shaped like it.

1. 14.1 · Domain of a vector function

Find the domain of $\mathbf{r}(t) = \left\langle \sqrt{t-1}, \ln(4-t), \frac{1}{t} \right\rangle$.

Answer: (A) $[1, 4)$

THE WORK

1. The domain is the **intersection** of the three component domains.
2. $\sqrt{t-1}$ needs $t-1 \geq 0$, so $t \geq 1$ — closed at 1.
3. $\ln(4-t)$ needs $4-t > 0$, so $t < 4$ — open at 4.
4. $\frac{1}{t}$ needs $t \neq 0$, which $t \geq 1$ already guarantees.
5. Intersecting: $1 \leq t < 4$, that is $[1, 4)$.

TRAP Making both ends the same. The square root allows equality, the logarithm does not — so one end is closed and the other is open.

2. 14.1 · Circle at a given height

Which function traces the circle of radius 3 centred on the z -axis in the plane $z = 5$, counterclockwise as seen from above?

Answer: (A) $\mathbf{r}(t) = \langle 3 \cos t, 3 \sin t, 5 \rangle$

THE WORK

1. Staying in the plane $z = 5$ forces the third component to be the constant 5.
2. Radius 3 centred on the axis gives $x = 3 \cos t$, $y = 3 \sin t$; then $x^2 + y^2 = 9$.
checkmark
3. $(+\cos, +\sin)$ runs counterclockwise viewed from $+z$ looking down.

TRAP Option B has $z = t$ — that is a helix, not a plane curve. Option D reverses the orientation. Option E shifts the **centre** to $(3, 3)$ and shrinks the radius to 1.

3. 14.1 · The surface a curve lies on

On which surface does $\mathbf{r}(t) = \langle t \cos t, t \sin t, t \rangle$ lie for $t \geq 0$?

Answer: (A) the cone $x^2 + y^2 = z^2$

THE WORK

1. Compute the combination the candidate surfaces care about:
2. $x^2 + y^2 = t^2 \cos^2 t + t^2 \sin^2 t = t^2(\cos^2 t + \sin^2 t) = t^2$.
3. And $z = t$, so $z^2 = t^2$. Therefore $x^2 + y^2 = z^2$ — the cone.
4. (The curve is a spiral climbing the cone, widening as it rises.)

TRAP Stopping at $x^2 + y^2 = t^2$ and matching it to z rather than z^2 . That would give the paraboloid, option C. Compare like with like.

4. 14.1 · Intersection with a surface

The helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ meets the sphere $x^2 + y^2 + z^2 = 5$. At which $t \geq 0$?

Answer: (A) $t = 2$

THE WORK

1. Substitute the components into the surface equation:

2. $\cos^2 t + \sin^2 t + t^2 = 5$.

3. $\cos^2 t + \sin^2 t = 1$ always, so $1 + t^2 = 5$, giving $t^2 = 4$ and $t = 2$ for $t \geq 0$.

TRAP Not using the Pythagorean identity and trying to solve a transcendental equation. On a helix the first two components always contribute exactly 1.

5. 14.2 · Derivative of a vector function

For $\mathbf{r}(t) = \langle t^2, e^{2t}, \ln t \rangle$, find $\mathbf{r}'(1)$.

Answer: (A) $\langle 2, 2e^2, 1 \rangle$

THE WORK

1. Differentiate componentwise: $\mathbf{r}'(t) = \left\langle 2t, 2e^{2t}, \frac{1}{t} \right\rangle$.

2. At $t = 1$: $\langle 2, 2e^2, 1 \rangle$.

TRAP Two chain-rule slips live here. $\frac{d}{dt}e^{2t} = 2e^{2t}$, not e^{2t} (option B), and $\frac{d}{dt}\ln t = \frac{1}{t}$, which is 1 at $t = 1$, not $\ln 1 = 0$ (option C).

6. 14.2 · Tangent line to a curve

Find parametric equations for the tangent line to $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ at the point $(2, 4, 8)$.

Answer: (A) $x = 2 + s, y = 4 + 4s, z = 8 + 12s$

THE WORK

1. First find the parameter value at that point. $x = t = 2$, and indeed $t^2 = 4, t^3 = 8$. So $t = 2$.

2. $\mathbf{r}'(t) = \langle 1, 2t, 3t^2 \rangle$, so $\mathbf{r}'(2) = \langle 1, 4, 12 \rangle$ — the direction.

3. Line through $(2, 4, 8)$ with direction $\langle 1, 4, 12 \rangle$: $x = 2 + s, y = 4 + 4s, z = 8 + 12s$.

TRAP Option C uses $\mathbf{r}'$ with t never substituted, effectively $\langle 1, 2, 3 \rangle$. Option B swaps the point and the direction. Option F evaluates $3t^2$ as $3 \cdot 2 = 6$ instead of $3 \cdot 4 = 12$.

7. 14.2 · Derivative rules

$\mathbf{r}(t)$ is differentiable and $|\mathbf{r}(t)|$ is **constant**. What is $\mathbf{r}(t) \cdot \mathbf{r}'(t)$?

Answer: (A) 0

THE WORK

1. Use the product rule for the dot product: $\frac{d}{dt}[\mathbf{r} \cdot \mathbf{r}] = \mathbf{r}' \cdot \mathbf{r} + \mathbf{r} \cdot \mathbf{r}' = 2\mathbf{r} \cdot \mathbf{r}'$.

2. But $\mathbf{r} \cdot \mathbf{r} = |\mathbf{r}|^2$, which is a constant, so its derivative is 0.

3. Hence $2\mathbf{r} \cdot \mathbf{r}' = 0$, so $\mathbf{r} \cdot \mathbf{r}' = 0$.

4. Geometrically: a curve of constant distance from the origin lies on a sphere, and its velocity must be tangent to that sphere — perpendicular to the radius.

TRAP This is the fact behind uniform circular motion: constant speed means the acceleration is perpendicular to the velocity, which is the same statement applied to $\mathbf{v}$ instead of $\mathbf{r}$.

8. 14.2 · Definite integral

Evaluate $\int_0^1 \langle 6t^2, 4t, e^t \rangle dt$.

Answer: (A) $\langle 2, 2, e - 1 \rangle$

THE WORK

1. Integrate componentwise: $\langle 2t^3, 2t^2, e^t \rangle$ evaluated from 0 to 1.

2. $x: 2(1)^3 - 0 = 2.$ $y: 2(1)^2 - 0 = 2.$ $z: e^1 - e^0 = e - 1.$

3. So the integral is $\langle 2, 2, e - 1 \rangle$.

TRAP Forgetting the lower limit on the exponential. $e^0 = 1$, not 0, so the third component is $e - 1$ — option B is the version that drops it.

9. 14.3 · Velocity, speed, acceleration

A particle has position $\mathbf{r}(t) = \langle 3t, t^2, t^3 \rangle$. Find its **speed** at $t = 1$.

Answer: (A) $\sqrt{22}$

THE WORK

1. $\mathbf{v}(t) = \mathbf{r}'(t) = \langle 3, 2t, 3t^2 \rangle$, so $\mathbf{v}(1) = \langle 3, 2, 3 \rangle$.

2. Speed is the **magnitude** of velocity: $|\mathbf{v}(1)| = \sqrt{9 + 4 + 9} = \sqrt{22}$.

TRAP Speed is a number, not a vector, and it comes from $\mathbf{v}$ — not from $\mathbf{r}$ and not from $\mathbf{a}$. Differentiating t^3 to t^2 instead of $3t^2$ gives $\sqrt{14}$, option B.

10. 14.3 · Circular motion

For $\mathbf{r}(t) = \langle 4 \cos 3t, 4 \sin 3t \rangle$, which statement is true?

Answer: (A) the speed is constant at 12, and the acceleration points toward the origin

THE WORK

1. $\mathbf{v} = \langle -12 \sin 3t, 12 \cos 3t \rangle$, so $|\mathbf{v}| = 12\sqrt{\sin^2 3t + \cos^2 3t} = 12$ — constant.

2. $\mathbf{a} = \langle -36 \cos 3t, -36 \sin 3t \rangle = -9\mathbf{r}(t)$.

3. A negative multiple of the position vector points straight back at the origin. (This is centripetal acceleration.)

4. Also $\mathbf{v} \cdot \mathbf{a} = 0$, so the acceleration is perpendicular to the velocity, never parallel to it.

TRAP “Constant speed” is not “no acceleration” — option F. The direction is changing the whole time, and changing direction is acceleration. Note also that the speed is 12, not the radius 4: the 3 inside the trig functions multiplies it.

11. 14.3 · Position from acceleration

A particle has acceleration $\mathbf{a}(t) = \langle 0, 2, 6t \rangle$, with $\mathbf{v}(0) = \langle 1, 0, 0 \rangle$ and $\mathbf{r}(0) = \langle 0, 1, 2 \rangle$. Find $\mathbf{r}(1)$.

Answer: (A) $\langle 1, 2, 3 \rangle$

THE WORK

1. Integrate once: $\mathbf{v}(t) = \langle 0, 2t, 3t^2 \rangle + \mathbf{C}_1$. Since $\mathbf{v}(0) = \langle 1, 0, 0 \rangle$, $\mathbf{C}_1 = \langle 1, 0, 0 \rangle$, so $\mathbf{v}(t) = \langle 1, 2t, 3t^2 \rangle$.

2. Integrate again: $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle + \mathbf{C}_2$. Since $\mathbf{r}(0) = \langle 0, 1, 2 \rangle$, $\mathbf{C}_2 = \langle 0, 1, 2 \rangle$.

3. $\mathbf{r}(t) = \langle t, t^2 + 1, t^3 + 2 \rangle$, so $\mathbf{r}(1) = \langle 1, 2, 3 \rangle$.

TRAP Two integrations mean **two** vector constants, and each is fixed by its own initial condition. Losing $\mathbf{C}_1$ drops the x motion entirely; losing $\mathbf{C}_2$ gives $\langle 1, 1, 1 \rangle$.

12. 14.3 · Projectile · time of flight

A ball is launched from a height of 15 m with initial velocity $\langle 20, 0, 10 \rangle$ m/s, under acceleration $\langle 0, 0, -10 \rangle$ m/s². When does it hit the ground?

Answer: (A) $t = 3$ s

THE WORK

1. Only the vertical component matters. Integrating twice, $z(t) = 15 + 10t - 5t^2$ (initial height, plus initial vertical velocity, minus $\frac{1}{2}gt^2$).

2. Set $z = 0$: $5t^2 - 10t - 15 = 0$, so $t^2 - 2t - 3 = 0$, that is $(t - 3)(t + 1) = 0$.

3. The positive root is $t = 3$ s.

TRAP Dropping the initial height and solving $10t - 5t^2 = 0$ gives $t = 2$ — option B. The launch height is what keeps it in the air longer. And the coefficient is $\frac{1}{2}g = 5$, not $g = 10$.

13. 14.3 · Projectile · maximum height

Same ball: launched from 15 m with velocity $\langle 20, 0, 10 \rangle$ m/s and acceleration $\langle 0, 0, -10 \rangle$ m/s². What is its maximum height?

Answer: (A) 20 m

THE WORK

1. The peak is where the **vertical velocity** is zero, not where $z = 0$.

2. $z'(t) = 10 - 10t = 0$ gives $t = 1$ s.

3. $z(1) = 15 + 10(1) - 5(1)^2 = 15 + 10 - 5 = 20$ m.

TRAP Answering 15 m, the launch height — the ball is still going up when it leaves. Or solving $z(t) = 0$, which finds the ground, not the peak.

7. Walking-in checklist

Say each of these out loud. Anything you cannot, go back to that problem.

- | | |
|---|--|
| ☐ Why differentiating goes slot by slot | ☐ Why $\vec{a} \perp \vec{v}$ at constant speed |
| ☐ Why $\vec{r}'$ points along the curve | ☐ The three projectile questions |
| ☐ The difference between velocity and speed | ☐ Which slot answers each of them |
| ☐ How to build $\vec{T}$, and why $ \vec{T} = 1$ | ☐ Domain = intersect every slot |
| ☐ Why $\vec{C}$ has three components | ☐ $\sqrt{\quad}$ closed, ln open |
| ☐ How $t = 0$ finds $\vec{C}$ | ☐ Arc length = rate $\times$ time, summed |

In the room. Read the question twice and ask *which object does it want* — a vector or a number? Then: differentiate, substitute, and only then take a length if it asked for speed or arc length. No calculator, so the numbers will be clean; if they are not, you have made an arithmetic slip.